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Post-Mortem Project Analysis: Failure of Apple AirPower

Background

On September 2017, Apple revealed the iPhone X alongside a new accessory, the AirPower charging mat. Most wireless chargers at the time were new and not that reliable, requiring users to position a single device precisely over a hidden coil. AirPower was marketed as a completely effortless fix. The promise was made that if you lay down an iPhone, an Apple Watch, and a pair of AirPods anywhere on the mat, all at once, without worrying, all devices would start charging and no need for cords

Apple originally targeted an early 2018 release, but that window came and went. Throughout the rest of the year, mentions of the accessory quietly started disappearing from marketing pages. Finally in March of 2019, Apple officially pulled the plug and admitted the product simply could not meet its standards. It has remained a rare, misstep in Apple's hardware management and a clear example of public marketing commitments of biting off more than they could chew or promise I should say.

Issues

- Heat Buildup: Packing in an overlapping mesh induction coil into an ultra-slim, shell trapped excessive amount of heat. Firing multiple coils at once threatened device battery health and constantly tripped the phones to shutoff.
- Electromagnetic Interference (EMI): Stacking dozens of coils running at different frequencies caused severe magnetic interference with each other. This warped charging fields and caused prototypes to fail FCC radio-frequency emission standards.
- Announcing Before Validating: Apple promised an early-2018 launch while the product was still in a delicate bench prototype. Bypassing standard stages like thermal testing and Design for Manufacturability (DFM), it left engineers boxed in by unverified public commitments.

- Over-Engineered Architecture: Rather than a simple charger, AirPower used custom silicon and a stripped-down iOS build to route power dynamically and push UI animations. This added unnecessary software bugs and glitches which led to the processor heating up in the phones body with zero space to dissipate.
- Aesthetics Over Physics: Design mandates demanded a paper-thin, silent mat without vents or fans. With active cooling ruled out, engineering was stuck between non-negotiable styling and the realities of multi-coil heat transfer.

Lessons learned

- Physics Trumps Styling: Aesthetics can't bypass basic thermodynamics. When hardware creates more heat than a sealed, slim case can dissipate, project managers need to renegotiate design specs early instead of forcing impossible goals on engineering.
- Hold Stage Gates Before Going Public: Never tie an unverified lab prototype to a public launch date. Clearing thermal checks, FCC testing, and manufacturing validation must happen before marketing locks in deadlines that trap the team.
- Simple Hardware Beats Complex Workarounds: Apple eventually nailed wireless alignment in 2020 with MagSafe. Rather than brute-forcing coverage with dozens of coils and microchips, a simple magnetic ring snapped devices right where they needed to be. Clean mechanical fixes almost always beat over-engineered electronic ones.
- Watch Out for Accessory Scope Creep: Keep peripherals focused on their actual job. Turning a charging mat into a mini-computer running an OS just to show lock-screen animations added needless failure points and extra heat for very little payoff.
- Set Hard Go/No-Go Milestones: If core technical hurdles stay unsolved cycle after cycle, leadership has to step in rather than fall for the sunk-cost fallacy. Defining clear kill criteria early saves engineering talent, cuts wasted time and protects brand credibility by not having to back track.